COVID-19 Status Prediction

A corrected machine learning evaluation using group-aware splitting, training-only random oversampling, and multi-metric model comparison.

5,434	20	3	95.24%
records	input features	models compared	best mean CV accuracy

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Educational use only. This project must not be used for medical diagnosis or clinical decision-making.

1. Project Overview

This project predicts the binary COVID-19 status recorded in a symptom-and-exposure dataset. It demonstrates an end-to-end classification workflow: data inspection, exploratory analysis, class-imbalance handling, leakage-aware splitting, model development, and evaluation.

Objectives

- Examine dataset quality, structure, missing values, duplicate patterns, and target imbalance.
- Explore symptom, health-condition, and exposure indicators associated with the recorded target.
- Prevent identical feature patterns from appearing in both training and test sets.
- Compare Logistic Regression, K-Nearest Neighbors (KNN), and Decision Tree classifiers.
- Evaluate performance with accuracy, balanced accuracy, precision, recall, F1-score, macro F1-score, and confusion matrices.

Dataset

Property	Description
Observations	5,434 records
Variables	20 input features and 1 target variable
Feature groups	Symptoms, health conditions, and exposure-related indicators
Target	COVID-19: No (0) or Yes (1)
Target distribution	No: 1,051 | Yes: 4,383
Missing values	0
Unique input patterns	453

The target is imbalanced: positive records are the majority. The dataset also contains many repeated binary feature patterns. These characteristics directly informed the splitting and evaluation strategy.

Exploratory Analysis

Feature prevalence and correlation were used as descriptive tools to identify associations within this dataset. They do not establish causation and should not be interpreted as clinical evidence.

2. Methodology

1	**Prepare**	Strip column-name whitespace and map binary Yes/No values to 1/0.
2	**Group**	Hash identical input-feature patterns so repeated patterns remain together.
3	**Split**	Use StratifiedGroupKFold to create group-separated training and holdout sets.
4	**Balance**	Apply RandomOverSampler to training data only; preserve the test distribution.
5	**Model**	Train Logistic Regression, KNN, and Decision Tree; select KNN k with training-only CV.
6	**Evaluate**	Assess untouched holdout performance and 10-fold group-aware cross-validation.

Data Leakage Prevention

Identical input patterns were assigned to the same group. The resulting holdout split contained zero shared feature patterns between training and testing.

The training set contained 4,402 records and the holdout test set contained 1,032 records. Random oversampling increased the minority class only within training from 776 to 3,626 records, matching the majority class. The test set remained unchanged (275 negative and 757 positive records).

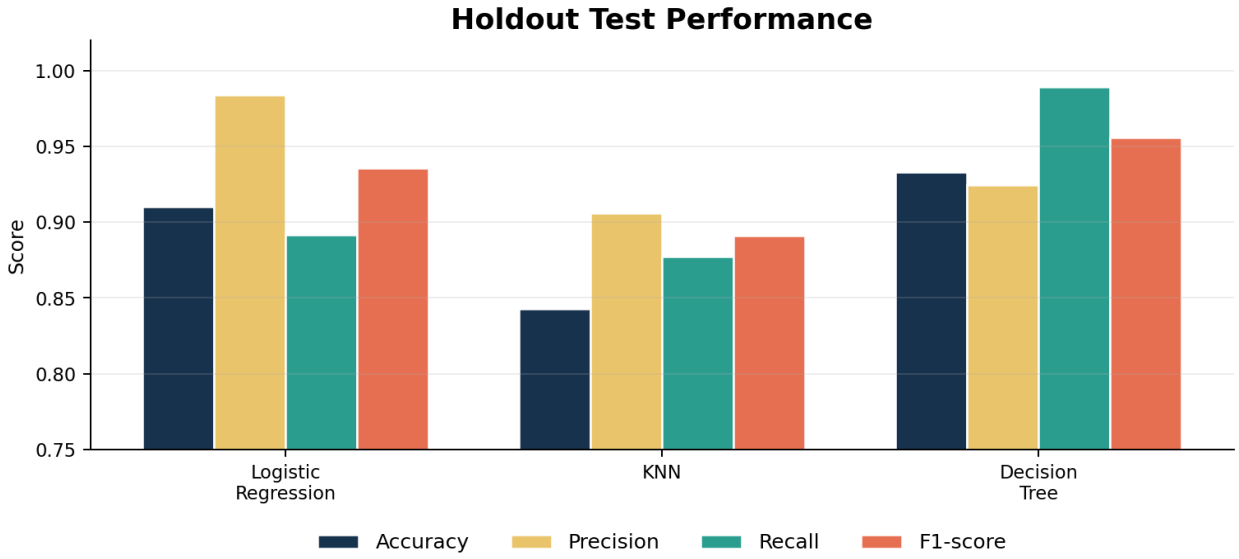
Models and Validation

- Logistic Regression: linear baseline with max_iter=1000.
- KNN: k=3 selected using five-fold group-aware cross-validation on training data.
- Decision Tree: nonlinear classifier with random_state=42.
- Final stability assessment: 10-fold stratified group cross-validation with oversampling inside each training fold.

3. Results

Holdout Test Results

Model	Accuracy	Precision	Recall	F1-score
Decision Tree	93.31%	92.47%	98.94%	95.60%
Logistic Regression	90.99%	98.40%	89.17%	93.56%
KNN	84.30%	90.59%	87.71%	89.13%



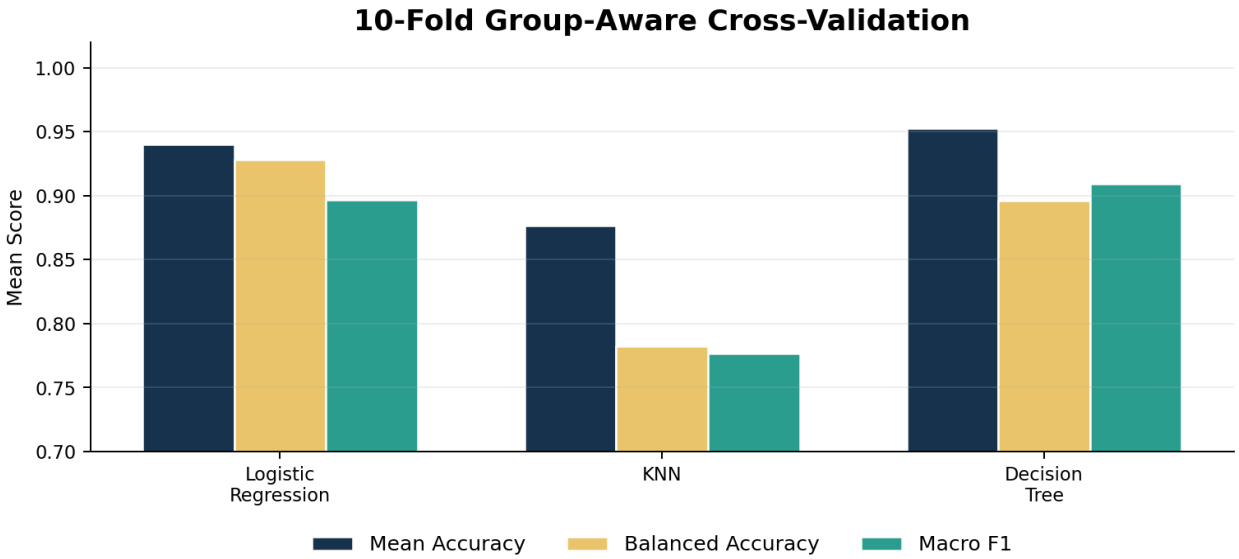
Decision Tree produced the strongest holdout accuracy, recall, and positive-class F1-score. It correctly identified 749 of 757 positive records, with 8 false negatives. Logistic Regression achieved the highest positive-class precision.

Holdout Confusion Matrix Counts

Model	TN	FP	FN	TP
Logistic Regression	264	11	82	675
KNN	206	69	93	664
Decision Tree	214	61	8	749

4. Cross-Validation and Model Selection

Model	Accuracy	Balanced Acc.	Precision	Recall	F1	Macro F1
Decision Tree	95.24%	89.61%	95.86%	98.25%	97.02%	90.91%
Logistic Regression	93.98%	92.80%	97.35%	95.05%	96.16%	89.67%
KNN	87.64%	78.23%	92.30%	92.95%	92.37%	77.64%



95.24%	92.80%	90.91%
Decision Tree mean accuracy	Logistic Regression balanced accuracy	Decision Tree macro F1

Selected model: Decision Tree, based on its overall accuracy, macro F1-score, and positive-class recall. Logistic Regression remains a strong alternative when balanced discrimination across classes is the primary objective.

Interpretation

The results are specific to this dataset and validation design. High performance does not demonstrate clinical validity. Repeated binary patterns, potential sampling bias, and limited contextual variables constrain generalizability.

5. Conclusions, Limitations, and Resources

Conclusion

The corrected workflow compared three classifiers while addressing class imbalance and reducing leakage from repeated input patterns. Decision Tree achieved the highest mean cross-validation accuracy (95.24%) and macro F1-score (90.91%), and reached 93.31% accuracy on the untouched group-separated holdout set. Logistic Regression achieved the highest mean balanced accuracy (92.80%).

Limitations

- The dataset contains binary indicators and limited demographic or temporal context.
- The source population may not represent other countries, time periods, or clinical settings.
- Some identical input patterns have conflicting target labels, indicating ambiguity or noise.
- Oversampling duplicates minority training records; it does not create new clinical evidence.
- No external clinical validation was performed.

Technology Stack

Data	Visualization	Machine Learning	Environment
Python, NumPy, Pandas	Matplotlib, Seaborn	Scikit-learn, Imbalanced-learn	Google Colab, GitHub

Project Resources

GitHub repository
Open the notebook in Google Colab
Dataset source on Kaggle

This report documents an educational machine learning exercise. Its outputs are not medical advice and must not be used to diagnose, treat, or make clinical decisions.

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